STA 5635 HW 6

Palmer Swanson

February 15, 2026

Problem 1

(1a) Two random forest classifiers. One with 22 features, the other with 500. Table in figure 1

Model	Train Misclass. Error	Test Misclass. Error
Random Forest, 22 Features	0.000000	0.296667
Random Forest, 500 Features	0.000000	0.156667

Table 1: Train and Test Misclassification Errors, part a

(1b) Compute feature importances using mean decrease in impurity and the permutation feature importances.

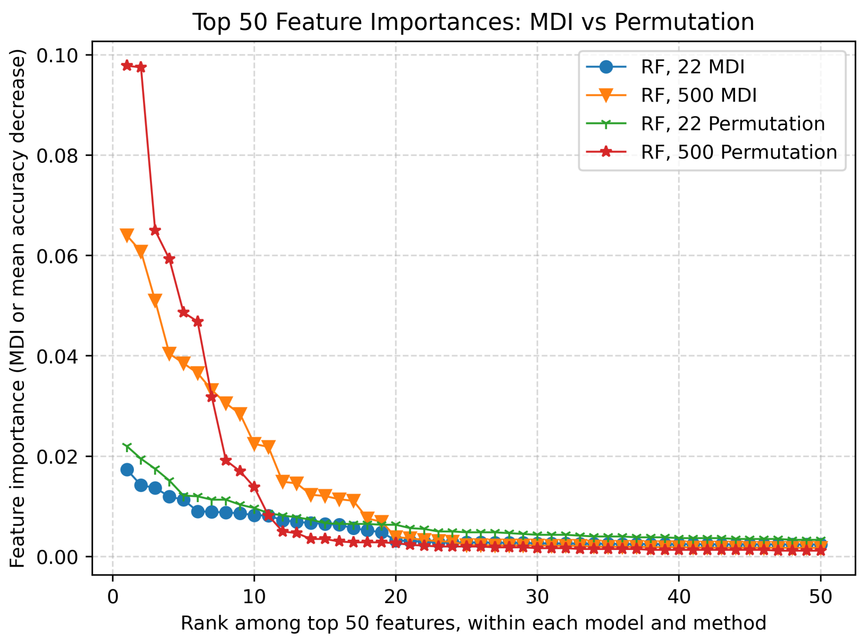

Figure 1: Feature Importances for the two RF, computed via MDI and via Permutations, part b.

(1c) Selecting $k \in \{10, 20, 30, 40\}$ most important features by the two methods on the two classifiers, train RF classifier. The test and train errors are reported in table 2.

Label	k	Train Misclass. Error	Test Misclass. Error
RF_22_MDI	10	0.000000	0.156667
RF_22_MDI	20	0.000000	0.118333
RF_22_MDI	30	0.000000	0.121667
RF_22_MDI	40	0.000000	0.128333
RF_500_MDI	10	0.000000	0.121667
RF_500_MDI	20	0.000000	0.118333
RF_500_MDI	30	0.000000	0.116667
RF_500_MDI	40	0.000000	0.131667
RF_22_PERM	10	0.000000	0.136667
RF_22_PERM	20	0.000000	0.135000
RF_22_PERM	30	0.000000	0.133333
RF_22_PERM	40	0.000000	0.135000
RF_500_PERM	10	0.000000	0.120000
RF_500_PERM	20	0.000000	0.123333
RF_500_PERM	30	0.000000	0.130000
RF_500_PERM	40	0.000000	0.131667

Table 2: Training and test misclassification errors for the fitted models, part c.

(1d) For $k \in \{10, 20, 30, 40\}$, two plots, one of each depth d. $d = 4$ is figure 2, $d = 5$ found in figure 3. Of the 320 classifiers, we can report the configuratio with the lowest test error in table: 3

Label	k	d	n_estimators	Train Misclass. Error	Test Misclass. Error
RF_500_PERM	10	5	80	0.031500	0.130000

Table 3: Configuration with smallest test error.

(1e) Of the models, the random forest classifier from part c using selecting the 30 most importance as determined by MDI where each split is from all 500 features has the lowest test error of 0.11667.

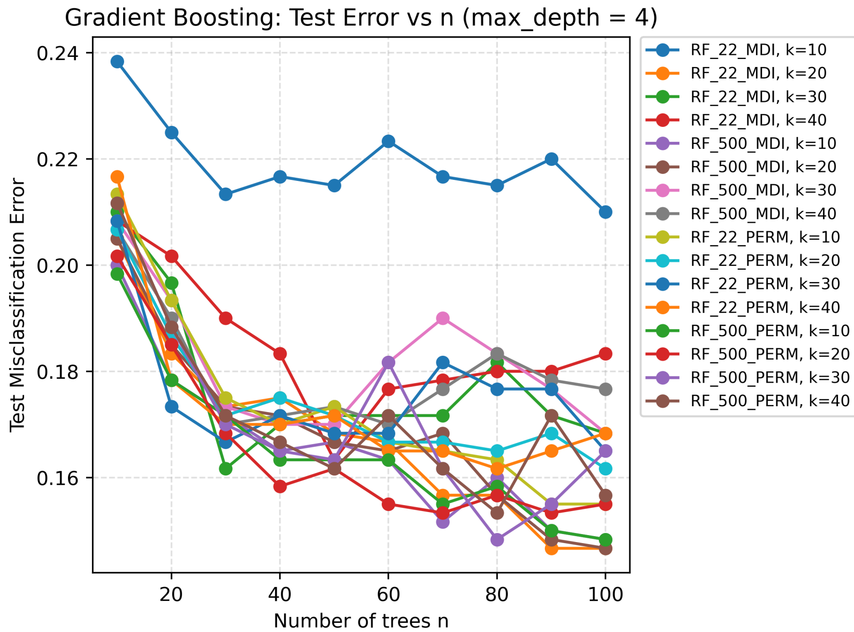

Figure 2: Test Errors vs. n for the 16 classifiers. d=4

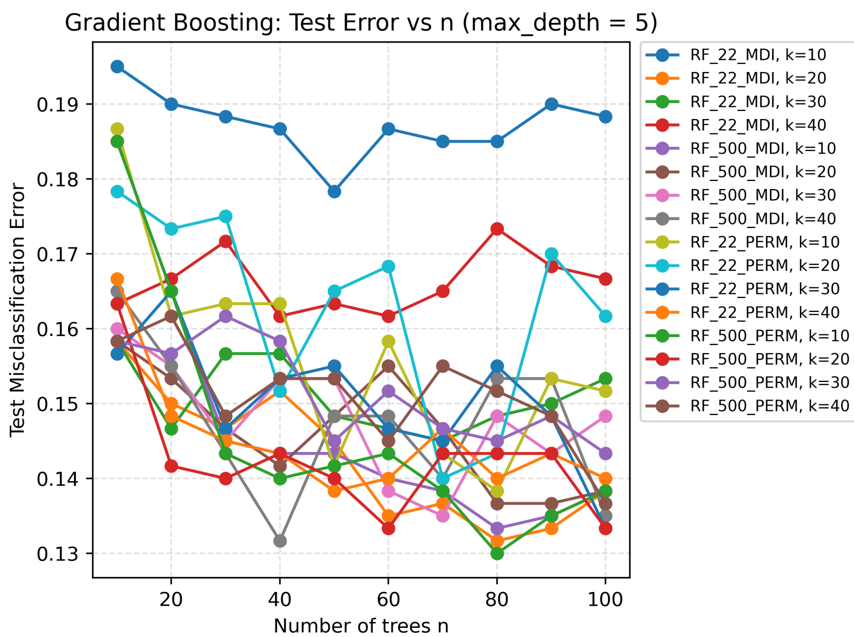

Figure 3: Test Errors vs. n for the 16 classifiers. d=5

Code